

DATA CLEANING

CLEAN DATA. BETTER INSIGHTS. SMARTER DECISIONS.



- IDENTIFY ISSUES
- CLEAN DATA
- IMPROVE ACCURACY
- DRIVE BETTER RESULTS



Raw data is never perfect.
Data Cleaning helps you remove errors, inconsistencies, duplicates, and missing values to make your data reliable and analysis-ready.

RAW DATA (UNSTRUCTURED)



Customer Name	Email	Phone	Revenue	Date	Status
rahul sharma	Rahul@Gmail.Com	98765-43210	45,000	1/5/24	completed
Neha Verma	neha.verma@gmail.com	9876543210	89,000	05-01-2024	Completed
Amit Kumar	AMITkumar@yahoo.com	98 765 43210	1,25,000	2024/01/05	COMPLETED
Priya Singh	priya_singh@gmail.co	9876543210	60,000	1-5-2024	Pending
rohan mehta	rohan.mehta@gmail.com	98765-43210	55,000	05/01/24	pending
Simran Kaur	simran.kaur@gmail.com	9876543210	70,000	5 Jan 2024	Cancelled
Vivek joshi	vivek@GMAIL.com	98765 43210	invalid	01-05-24	CANCELLED
Karan Malhotra	karan.malhotra@yahoo.com	9876543210	95,000	5/1/24	Completed
Pooja Nair	pooja.nair@gmail.com	-	45,000	2024-01-05	pending

CLEAN DATA (STRUCTURED)



Customer Name	Email	Phone	Revenue	Date	Status
Rahul Sharma	rahul@gmail.com	9876543210	45000	05-01-2024	Completed
Neha Verma	neha.verma@gmail.com	9876543210	89000	05-01-2024	Completed
Amit Kumar	amitkumar@yahoo.com	9876543210	125000	05-01-2024	Completed
Priya Singh	priya_singh@gmail.com	9876543210	60000	05-01-2024	Pending
Rohan Mehta	rohan.mehta@gmail.com	9876543210	55000	05-01-2024	Pending
Simran Kaur	simran.kaur@gmail.com	9876543210	70000	05-01-2024	Cancelled
Vivek Joshi	vivek@gmail.com	9876543210	0	05-01-2024	Cancelled
Karan Malhotra	karan.malhotra@yahoo.com	9876543210	95000	05-01-2024	Completed
Pooja Nair	pooja.nair@gmail.com	0	45000	05-01-2024	Pending

COMMON DATA QUALITY ISSUES



Inconsistent Text

Different letter cases, extra spaces



Wrong Formats

Phone numbers, dates, emails in different formats



Numbers as Text

Numbers stored as text with symbols or commas



Missing Values

Blank, NULL, N/A or incomplete data



Duplicates

Repeated records or duplicate entries



Invalid Emails

Incorrect or incomplete email addresses



Inconsistent Categories

Different spellings for the same value



Special Characters

Unwanted symbols, line breaks, extra chars

WHAT DATA CLEANING HELPS YOU DO

- ✓ Remove errors and inconsistencies
- ✓ Standardize formats and values
- ✓ Handle missing and duplicate data
- ✓ Convert text to numbers or dates
- ✓ Improve data accuracy and reliability
- ✓ Prepare data for analysis and dashboards
- ✓ Save time and improve efficiency

POWERFUL EXCEL TOOLS & FUNCTIONS

TRIM

Remove extra spaces

PROPER UPPER LOWER

Standardize text case

LEFT RIGHT MID

Extract specific information

FIND SEARCH LEN

Locate text & calculate length

SUBSTITUTE REPLACE

Remove or replace text

TEXT VALUE

Convert text to numbers / dates



Flash Fill Text to Columns Remove Duplicates

Quick cleaning tools

“ CLEAN DATA IS THE FOUNDATION OF POWERFUL ANALYTICS.”



Data Cleaning in Excel

Module Objective

- Understand why data cleaning is one of the most important responsibilities of a Data Analyst.
- Identify dirty and inconsistent data.
- Clean datasets using Excel functions already learned.
- Combine multiple functions to solve real business problems.
- Prepare raw business data for dashboards, Pivot Tables, SQL, and Power BI.
- Solve interview-oriented data cleaning scenarios.

Module 1 – Introduction to Data Cleaning

What is Data Cleaning?

Data Cleaning is the process of identifying, correcting, and standardizing incorrect, incomplete, duplicate, inconsistent, or improperly formatted data before analysis.

Definition

Data Cleaning ensures that the dataset is accurate, complete, consistent, and ready for business analysis.

Why is Data Cleaning Important?

Nearly **70–80%** of a Data Analyst's time is spent cleaning and preparing data before performing analysis.

Dirty data leads to:

- Incorrect dashboards
- Wrong KPIs
- Duplicate customers
- Revenue calculation errors
- Poor business decisions

Business Example

Raw Data

Customer Name	City	Revenue
Rahul Sharma	Delhi	50000
rahul sharma	Delhi	50000
Rahul Sharma	Delhi	50000

Without cleaning:

Customer Count = 3

Actual Customer Count = 1

Business Benefits

- Accurate Reports
- Reliable Dashboards
- Better Decision Making
- Improved Forecasting
- Faster Analysis



Module 2 – Common Data Quality Problems

Data Analysts frequently encounter:

- Extra Spaces
- Wrong Letter Case
- Duplicate Records
- Missing Values
- Mixed Date Formats
- Numbers Stored as Text
- Special Characters
- Invalid Email IDs
- Blank Rows
- Inconsistent Categories

Module 3 – Cleaning Spaces (TRIM)

Note: The **TRIM** function was introduced in the Text Functions module. Here we apply it to solve a real business problem.

Business Problem

Customer Name

Rahul Sharma

Rahul Sharma

Rahul Sharma

All three appear different because of hidden spaces.

Formula

=TRIM(B2)

Business Use Cases

- Customer Names
- Employee Names
- Product Names
- Cities
- Vendor Names

Module 4 – Standardizing Letter Case

Functions already covered:

- UPPER
- LOWER
- PROPER

Now apply them in Data Cleaning.

Business Example

Raw Data

rahul SHARMA



<https://www.linkedin.com/in/the-venom/>

Shivansh

RAHUL sharma

Rahul Sharma

Formula

=PROPER(B2)

Result

Rahul Sharma

Business Use Cases

- Employee Names
- Customer Names
- City Names
- Product Categories

Module 5 – Removing Unwanted Characters

Functions already covered:

- SUBSTITUTE
- REPLACE

Business Example

Phone Number

98765-43210

Formula

=SUBSTITUTE(C2,"-","")

Result

9876543210

Business Uses

- Mobile Numbers
- GST Numbers
- PAN Numbers
- Invoice Numbers

Module 6 – Extracting Information

Previously Covered Functions

- LEFT
- RIGHT
- MID
- FIND
- SEARCH

Now combine them.



Example 1

Extract Email Username

Email

rahul@gmail.com

Formula

=LEFT(C2,FIND("@",C2)-1)

Result

rahul

Example 2

Extract Email Domain

Formula

=MID(C2,FIND("@",C2)+1,LEN(C2))

Result

gmail.com

Business Uses

- Email Analysis
- Customer Domains
- Company Email Extraction

Module 7 – Combining Functions

This is where interview questions are commonly asked.

Scenario 1

Extract First Name

Formula

=LEFT(B2,SEARCH(" ",B2)-1)

Scenario 2

Extract Last Name

Formula

=RIGHT(B2,LEN(B2)-SEARCH(" ",B2))

Scenario 3

Clean Name

Formula

=PROPER(TRIM(B2))

Scenario 4

Extract Domain

Formula

=RIGHT(C2,LEN(C2)-SEARCH("@",C2))

Scenario 5



<https://www.linkedin.com/in/the-venom/>

Shivansh

Remove Spaces + Convert Proper Case

Formula

=PROPER(TRIM(A2))

Scenario 6

Remove Hyphen

Formula

=SUBSTITUTE(D2,"-","")

Scenario 7

Extract Product Code

Product ID

LAP-10025

Formula

=LEFT(A2,SEARCH("-",A2)-1)

Result

LAP

Scenario 8

Extract Numeric ID

Formula

=RIGHT(A2,LEN(A2)-SEARCH("-",A2))

Module 8 – VALUE Function

Already covered.

Business Use.

Revenue

"45,000"

↓

Convert into Number.

Formula

=VALUE(SUBSTITUTE(H2,"",""))

Business Uses

- Imported CSV
 - ERP Data
 - SAP Reports
-

Module 9 – Remove Duplicates

Location

Data

↓

Remove Duplicates



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Shivansh

Business Uses

- Duplicate Customers
- Duplicate Orders
- Duplicate Products

Best Practice

Always keep a backup before removing duplicates.

Module 10 – Find & Replace

Shortcut

Ctrl + H

Business Uses

Replace

Pending

↓

In Progress

Replace

N/A

↓

Blank

Module 11 – Go To Special

Shortcut

F5

or

Ctrl + G

↓

Special

Business Uses

Find

- Blank Cells
- Formula Cells
- Constants
- Comments

Module 12 – Flash Fill

Shortcut

Ctrl + E



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Shivansh

Business Uses

- Extract
- First Name
- Last Name
- Email Username
- Phone Formatting
- Invoice Formatting

Module 13 – Text to Columns

Location

Data



Text to Columns

Business Uses

Split

Rahul Sharma



First Name

Last Name

Split

Email



Username



Domain



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Module 14 – Data Validation During Cleaning

Prevent future errors.

Examples

- Department Dropdown
- Gender Dropdown
- Status Dropdown

Business Uses

Prevents inconsistent spelling.

Module 15 – Cleaning Workflow Used by Data Analysts

1. Create Backup
2. Remove Blank Rows
3. Standardize Text
4. Remove Extra Spaces



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Shivansh

5. Convert Numbers Stored as Text
6. Standardize Dates
7. Remove Duplicates
8. Validate Data
9. Check Missing Values
10. Verify Final Dataset

Excel Shortcuts for Data Cleaning

Shortcut	Purpose
Ctrl + H	Find & Replace
Ctrl + E	Flash Fill
Ctrl + G / F5	Go To Special
Ctrl + T	Convert Range to Table
Ctrl + Shift + L	Apply/Remove Filters
Ctrl + Arrow Keys	Navigate Data
Ctrl + Space	Select Column
Shift + Space	Select Row
Ctrl + A	Select Entire Dataset
Alt + A + M	Remove Duplicates
Alt + A + E	Text to Columns

Business Case Scenarios

Case Study 1

Your company imports customer data from an ERP system.

Problems:

- Mixed cases
- Extra spaces
- Duplicate customers
- Blank phone numbers

How would you clean the data?

Case Study 2

Revenue is stored as

₹45,000

₹89,000

₹1,25,000

How will you convert it into numeric values for analysis?

Case Study 3

Extract usernames from company email addresses.

Case Study 4

Separate product code and numeric ID from:

LAP-1054

MON-3025

CHR-4520

Case Study 5

Standardize all customer names into Proper Case while removing unnecessary spaces.

Interview-Oriented Questions (with Answers)

Q1. Why is Data Cleaning important?

Answer: It improves data accuracy, removes inconsistencies, prevents incorrect analysis, and ensures reliable business reporting.

Q2. Which text functions are most commonly used for data cleaning?

Answer: TRIM, PROPER, UPPER, LOWER, LEFT, RIGHT, MID, FIND, SEARCH, SUBSTITUTE, REPLACE, VALUE, LEN.

Q3. Difference between FIND and SEARCH?

Answer: FIND is case-sensitive and does not support wildcards, whereas SEARCH is case-insensitive and supports wildcards.

Q4. How would you extract the username from an email?

Answer:

=LEFT(A2,FIND("@",A2)-1)

Q5. How would you extract the email domain?

Answer:

=MID(A2,FIND("@",A2)+1,LEN(A2))

Q6. Which function removes extra spaces?

Answer: TRIM()

Q7. How would you standardize names like "rAhUL sHarMa"?

Answer:

=PROPER(TRIM(A2))

Q8. Difference between REPLACE and SUBSTITUTE?

Answer: REPLACE changes characters based on their position, while SUBSTITUTE replaces specific text occurrences.



Q9. What shortcut opens Find & Replace?

Answer: Ctrl + H

Q10. What shortcut performs Flash Fill?

Answer: Ctrl + E

Q11. How do you remove duplicate records in Excel?

Answer: Data → Remove Duplicates (Shortcut: Alt + A + M)

Q12. How would you clean the provided dataset?

Expected approach:

1. Remove extra spaces using TRIM().
 2. Standardize names with PROPER().
 3. Convert emails to lowercase using LOWER().
 4. Remove hyphens/spaces from phone numbers using SUBSTITUTE().
 5. Convert revenue text to numbers using VALUE() and SUBSTITUTE().
 6. Standardize status values (Completed, Pending, Cancelled).
 7. Identify and remove duplicates.
 8. Fill or flag missing values.
 9. Convert dates into a consistent format.
 10. Validate the cleaned dataset before analysis.
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Practical Assignment (Based on the Provided Dataset)

Using the **Lecture_09_Data_Cleaning_Unstructured_Dataset.xlsx**, complete the following tasks:

1. Convert all customer names to **Proper Case**.
2. Remove all leading and trailing spaces from text fields.
3. Convert all email addresses to **lowercase**.
4. Remove spaces and hyphens from phone numbers.
5. Identify records with missing Customer IDs.
6. Standardize all Status values to one format (Completed, Pending, Cancelled).
7. Convert Revenue from text to numeric values.
8. Highlight invalid email addresses (missing @).
9. Remove duplicate customer records.
10. Prepare the dataset so it is ready for Pivot Tables, dashboards, SQL import, and Power BI.

